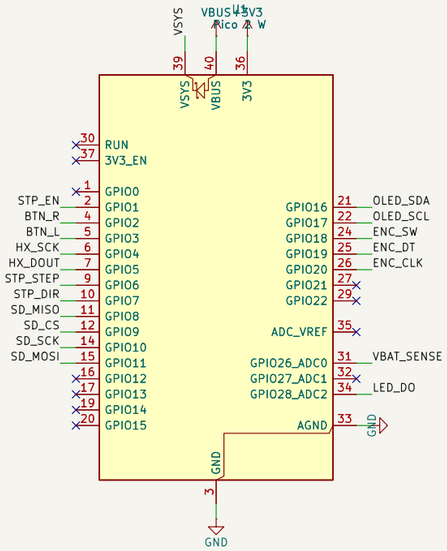
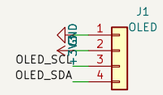
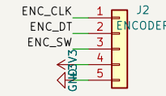
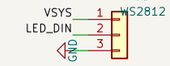
MixMeasureBuddy carrier board — Raspberry Pi Pico 2 W

Pin mapping = src/firmware/src/config.py (OLED I2C0 GP16/17, buttons GP3/GP2, HX711 GP4/GP5, encoder GP20/19/18, WS2812 GP28, SD SPI1 GP8–11). Buttons use the internal pull-ups.

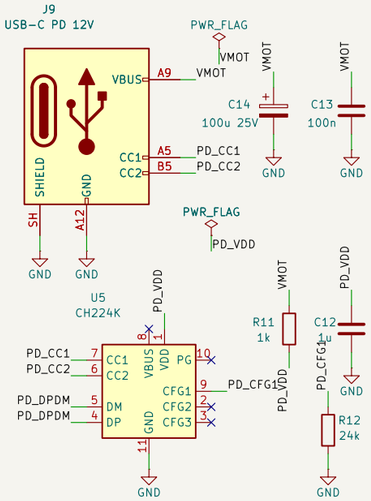




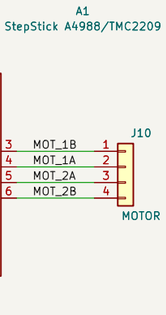


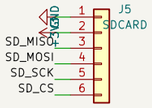


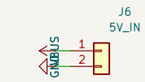


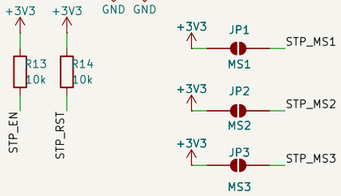


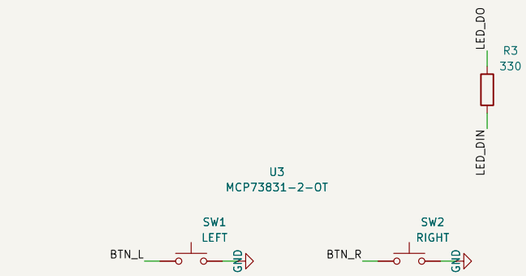
USB-C PD MOTOR SUPPLY (VMOT, separate from the Pico supply)
CH224K requests 12 V. R12: 6.8k=9 V, 24k=12 V, 56k=15 V
PD only: DP/DM shorted, not wired to the receptacle.

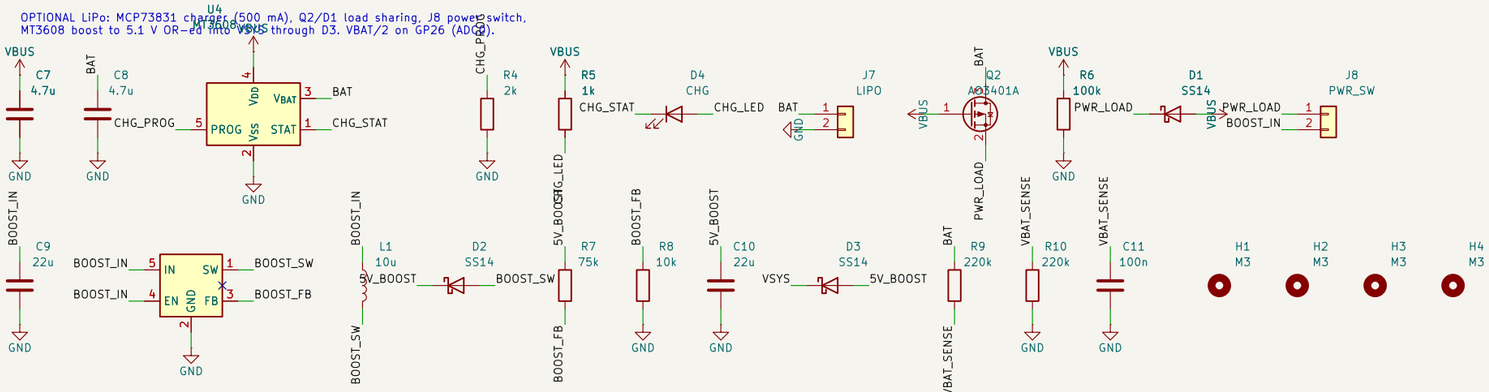












HX711: ready-made module plugs into J4 (1 = GND, 2 = DT, 3 = SCK, 4 = VCC 3V3).
Wire the load cell to the module: E+ red, E− black, A− white, A+ green.



STEPPER (StepStick socket): STEP = GP6, DIR = GP7,
EN = GP1 (active low, R13 pull-up keeps it off at boot).
RST/SLP pulled high. MS1–MS3 open = low, close JP1–JP3
to pull high (microstep mode depends on the module).
Coil 1 = J10 pins 1/2 (1B/1A), coil 2 = pins 3/4 (2A/2B).

Sheet: /	
File: mmb_carrier.kicad_sch	
Title: MixMeasureBuddy Pico 2 W carrier	
Size: A3	Date: 2026-10-31
KiCad E.D.A. 10.0.7	**Rev: 1.0**
	Id: 1/1